Experiment 22: Reference Resolution in Text using Python

1. Aim

To write a Python program that performs pronominal reference resolution within a text.

2. Procedure

1. Define a dictionary of entities and their gender attributes along with pronoun mappings.
2. Define a sample text containing named entities and pronouns.
3. Iterate through the words in the text and track the latest antecedent entity.
4. Resolve each pronoun to its matching antecedent noun.
5. Display the resolved pronoun-entity pairs.

3. Program

```
entities = {"John": "M", "Mary": "F"}
pronouns = {"he": "M", "she": "F"}

text = "John met Mary and she greeted him while he smiled"
context = {}

for w in text.split():
    if w in entities: context[entities[w]] = w
    elif w in pronouns: print(f"Pronoun '{w}' refers to:", context[pronouns[w]])
```

4. Output

```
Pronoun 'she' refers to: Mary
Pronoun 'he' refers to: John
```

5. Result

Thus, the Python program was successfully executed to perform reference resolution of pronouns within a text.